



LOGICAL DATA MODEL (LDM)

Document 03 of 09 · B2B Data Engineering Platform



Describes the business entities, attributes, relationships and history rules of the analytical model, independent of any database technology.

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Document Control

Revision History

Version	Date	Author	Description of Change
0.1	-	Syed Kazim Raza	Template created
1.0	29/09/2026	Syed Kazim Raza	Content populated: entities, attributes, relationships, history rules, logical-to-physical mapping
1.1	30/09/2026	Syed Kazim Raza	Related document paths updated (no model change)
1.2	04/10/2026	Syed Kazim Raza	LDM diagram redrawn (light theme, draw.io); note on business-rule IDs added in section 9
1.3	05/10/2026	Syed Kazim Raza	Formatting to documentation standard: table column widths, numbered figure and table captions with lists, figure descriptions, complete acronym list

Approvals

Role	Name	Signature	Date
Prepared by	Syed Kazim Raza		
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Related Documents

Ref	Document	Location
R1	Business Requirements Document (BRD)	Provided by DataRopes
R2	Project Document (technical reference)	docs/deliverables/project_document.md
R3	Project Plan	docs/deliverables/project_plan.md
R4	README / Deployment Guide	README.md

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1. Introduction

1.1 Purpose

This document defines the logical data model that the warehouse implements.

1.2 Scope

In scope:

- Business entities, attributes, keys, relationships and cardinality
- History (SCD) and business rules at the logical level

Out of scope:

- Data types, indexes, partitions (see PDM)

1.3 Intended Audience

Table 1: Intended Audience

Audience	How they use this document
Data Modellers / Engineers	Build the physical model from it
Analysts / Business	Confirm entities and definitions

1.4 Definitions & Acronyms

Table 2: Definitions & Acronyms

Term	Definition
AK	Alternate Key: a second unique identifier of an entity besides its primary key.
Attribute	A fact that describes an entity (e.g. order status).
B2B	Business-to-Business: companies selling to other companies.
BI	Business Intelligence: reporting and dashboards, here Power BI.
BR	Business Rule: a rule the data must follow, numbered BR-1 ... in the Source System Analysis.
BRD	Business Requirements Document: here the Project Requirements Document provided by DataRopes.
Conformed dimension	A dimension shared with the same meaning by several facts (e.g. Date).
DEP	Data Engineering Platform: prefix of the document IDs (B2B-DEP-01 ...).
Entity	A business object the organisation wants to keep information about (e.g. Order).
ERD	Entity-Relationship Diagram: a diagram of tables (entities) and their relationships.
Fact / Dimension	Kimball terms: facts hold measurable events, dimensions hold the descriptive context.
FK	Foreign Key: a column that references the primary key of another table.
Grain	What exactly one row (instance) represents.
HTTP	Hypertext Transfer Protocol: the web protocol recorded in the web logs (GET, POST ...).
ID	Identifier.
Identifier (PK)	Attribute(s) that uniquely identify one instance of an entity. Logical identifiers are business keys.
KPI	Key Performance Indicator: a business measure shown on the dashboard (KPI-01 ... KPI-15).
LDM	Logical Data Model (deliverable 03).
OLTP	Online Transaction Processing: an operational database such as the SQL Server source.

Term	Definition
PDM	Physical Data Model (deliverable 04).
SCD	Slowly Changing Dimension: SCD1 overwrites changes, SCD2 keeps every version with validity dates.
UUID	Universally Unique Identifier: the 32-character IDs used as business keys.
YYYYMMDD	Date written as year, month, day (e.g. 20260930); the format of date_key.

2. Modelling Approach

The warehouse follows the **Kimball dimensional approach**: business processes are modelled as facts at the lowest useful grain, surrounded by descriptive dimensions. This keeps queries simple for Power BI (star joins) and makes KPIs additive.

Because the platform covers **four business processes** (ordering, order lines, lead generation and web traffic), the model is a **galaxy schema (fact constellation)**: four fact entities share conformed dimensions instead of four isolated stars.

- **Date** is conformed across all four facts, so revenue, leads and traffic can be compared on the same calendar.
- **Customer, Company, Product and Supplier** are shared by Order and Order Line, so header-level and line-level KPIs slice identically.
- **Category** is modelled as its own logical entity but flattened into Product physically (it has one attribute and 10 instances).
- **Supplier Offer** resolves the many-to-many relationship between Supplier and Product and carries the offer's price and lead time.

Why this design: the source is a normalised OLTP model (source ERD) that is efficient for writing but needs many joins for analysis. The dimensional model trades some redundancy for fewer joins, stable keys and history tracking where the business needs it (Company, SCD Type 2).

Order and Marketing Lead are linked through the business key lead_id rather than a dimension, because a lead is itself an event (fact) with its own grain.

3. Subject Areas

Table 3: Subject Areas

Subject Area	Entities	Business Questions Answered
Sales	Order, Order Line, Date	Monthly revenue trend; revenue by company and product; gross margin (KPI 1-4)
Customer & Company	Customer, Company	Customer lifetime value; customer activity; sales by geography (KPI 5-7)
Product & Supply	Product, Category, Supplier, Supplier Offer	Supplier revenue; supplier-product performance, preferred suppliers (KPI 13-14)
Marketing	Marketing Lead, Order, Date	Lead conversion funnel; lead quality vs order value (KPI 11-12)
Web Activity	Web Request, Date	Traffic by device; geographic web traffic; error / bot share (KPI 8-10)

4. Logical Data Model Diagram

Figure 1 shows the logical model: business entities, their identifiers and the cardinality of each relationship, without technical data types.

Logical Data Model (LDM) — business entities and relationships

Technology-independent: business names, keys and cardinality, no data types. Normalised here; the physical model denormalises it into a galaxy schema.

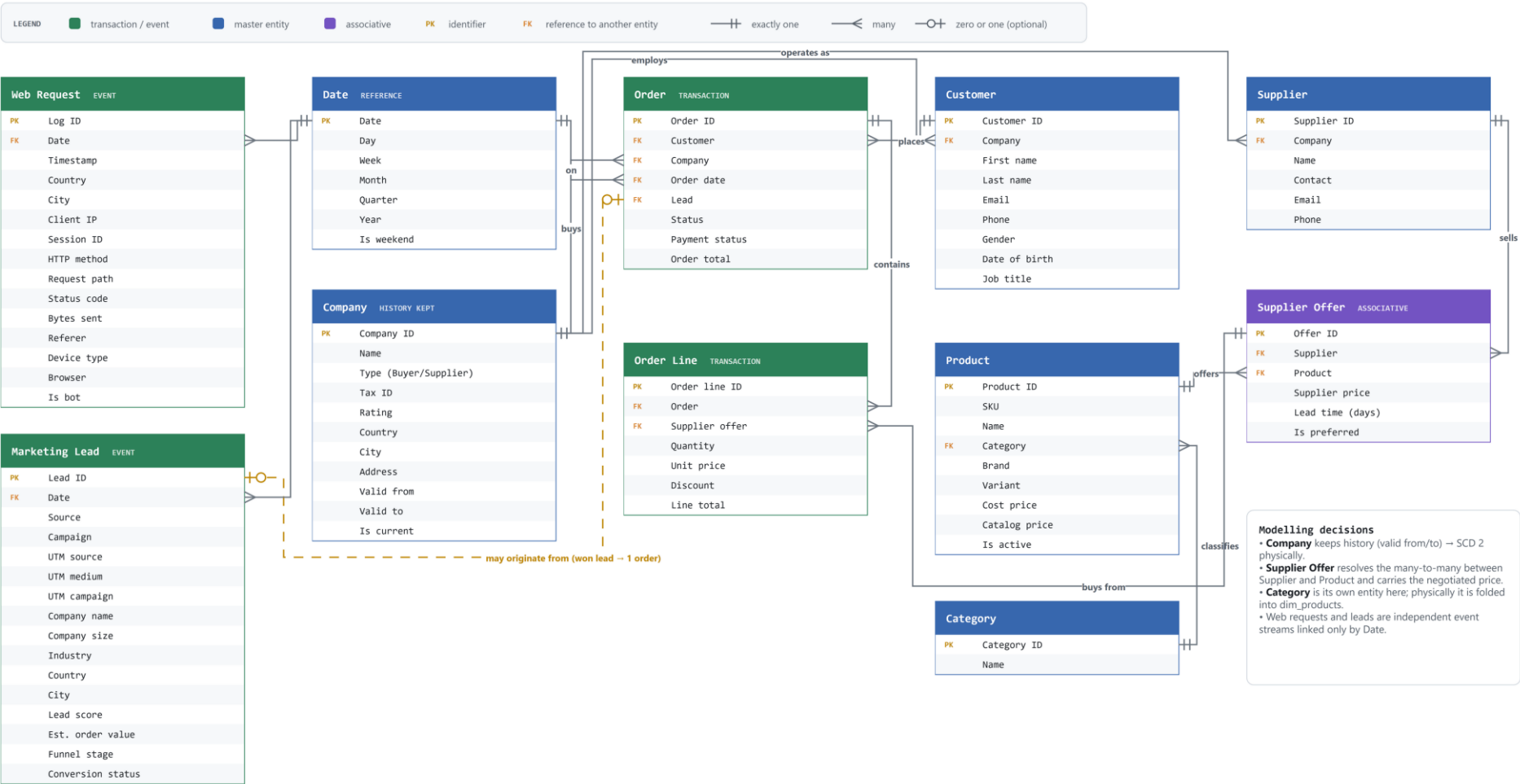


Figure 1: Logical data model — entities and relationships

5. Entity Definitions

Table 4: Entity Definitions

Entity	Type	Definition	Primary Identifier	Grain
Date	Dimension (reference)	A calendar day used to analyse all events.	Date key (YYYYMMDD)	One per calendar day
Company	Dimension (SCD2)	An organisation registered on the platform, acting as Buyer or Supplier.	company_id (+ validity period)	One per company version
Customer	Dimension	A person at a buyer company who places orders.	customer_id	One per customer
Supplier	Dimension	The selling account of a supplier company.	supplier_id	One per supplier
Category	Dimension (flattened)	A grouping of products.	category_id	One per category
Product	Dimension	An item that can be bought on the platform.	product_id (alt: sku)	One per product
Supplier Offer	Dimension / bridge	A product as offered by one supplier, with its price and lead time.	supplier_product_id (alt: supplier + product)	One per supplier-product pair
Order	Fact (transaction)	A purchase placed by a customer on behalf of a buyer company.	order_id	One per order
Order Line	Fact (transaction)	One supplier offer bought within an order, with quantity, price and discount.	order_item_id	One per order line
Marketing Lead	Fact (accumulating)	A prospect captured by a marketing campaign, tracked through the sales funnel.	lead_id	One per lead
Web Request	Fact (event)	One HTTP request made to the B2B web portal.	log_id	One per request

6. Attribute Definitions

The key attributes of each entity (49 of 100) in business language: identifiers, references and the attributes used by the KPIs. The complete list with data types is in 04b_Data_Dictionary.xlsx. Key: PK = identifier, FK = reference to another entity, AK = alternate (unique) key.

Table 5: Attribute Definitions

Entity	Attribute	Definition	Key	Domain / Allowed Values	Mandatory
Date	date key	Date surrogate key in YYYYMMDD format (FK to dim_date); also the partition key on facts.	PK		Yes

Entity	Attribute	Definition	Key	Domain / Allowed Values	Mandatory
Company	company id	Business identifier of the company (32-char hex UUID assigned by the Python generator).	PK		Yes
Company	company name	Legal/trading name of the company.	AK		Yes
Company	company type	Role of the company in the marketplace: Buyer or Supplier.		Buyer, Supplier	Yes
Company	cuit tax id	Company tax identification number (unique per company).	AK		Yes
Company	rating	Company rating on a 1.0-5.0 scale.		1.0 - 5.0	Yes
Company	effective start date	Start of validity of this company version.			Yes
Company	effective end date	End of validity (empty = current).			No
Company	is current	Marks the current version.			Yes
Customer	customer id	Business identifier of the customer (a buyer contact person).	PK		Yes
Customer	company id	Company the record belongs to.	FK Company		Yes
Customer	email	E-mail address (lower-cased from intermediate onwards).	AK		Yes
Supplier	supplier id	Business identifier of the supplier.	PK		Yes
Supplier	company id	Company the record belongs to.	FK Company		Yes
Supplier	email	E-mail address (lower-cased from intermediate onwards).	AK		Yes
Category	category id	Business identifier of the product category.	PK		Yes
Category	category name	Name of the product category.	AK		Yes
Product	product id	Business identifier of the product.	PK		Yes
Product	sku	Stock-keeping unit code of the product.	AK		Yes
Product	category id	Business identifier of the product category.	FK Category		Yes
Product	cost price	Unit cost price of the product.		>= 0	Yes
Product	catalog price	Unit list (catalog) price of the product; >= cost price.		>= cost price	Yes
Supplier Offer	supplier product id	Business identifier of a supplier-product offer (which supplier sells which product).	PK		Yes

Entity	Attribute	Definition	Key	Domain / Allowed Values	Mandatory
Supplier Offer	supplier id	Business identifier of the supplier.	FK Supplier		Yes
Supplier Offer	product id	Business identifier of the product.	FK Product		Yes
Supplier Offer	supplier price	Price at which the supplier offers the product.		>= 0	Yes
Order	order id	Business identifier of the order.	PK		Yes
Order	customer id	Business identifier of the customer (a buyer contact person).	FK Customer		Yes
Order	company id	Company the record belongs to.	FK Company		Yes
Order	lead id	Identifier of the marketing lead that generated the record (NULL if none).	FK Marketing Lead		No
Order	order date	Date and time the order was placed.	FK Date		Yes
Order	order status	Fulfilment status of the order.		Pending, Confirmed, Processing, Shipped, Delivered, Cancelled	Yes
Order	payment status	Payment status of the order.		pending, paid, failed, refunded	Yes
Order	order total	Total order amount (sum of line totals).		>= 0	Yes
Order Line	order item id	Business identifier of the order line.	PK		Yes
Order Line	order id	Business identifier of the order.	FK Order		Yes
Order Line	supplier product id	Business identifier of a supplier-product offer (which supplier sells which product).	FK Supplier Offer		Yes
Order Line	quantity	Quantity ordered on the line (> 0).		> 0	Yes
Order Line	unit price	Unit selling price on the line.		>= 0	Yes
Order Line	line total	Line amount = quantity x unit_price - discount_amount.		>= 0	Yes
Marketing Lead	lead id	Identifier of the marketing lead that generated the record (NULL if none).	PK		Yes
Marketing Lead	lead score	Lead quality score, 1-100.		1 - 100	Yes

Entity	Attribute	Definition	Key	Domain / Allowed Values	Mandatory
Marketing Lead	funnel stage	Current sales-funnel stage of the lead.		New, Contacted, Qualified, Proposal, Negotiation, Won, Lost	Yes
Marketing Lead	created at	Date and time the lead was captured (lead date).	FK Date		Yes
Marketing Lead	conversion status	Converted if at least one order references the lead, otherwise Not Converted.		Converted, Not Converted	Yes
Web Request	log id	Identifier of the web request log entry.	PK		Yes
Web Request	log timestamp	Timestamp of the web request.	FK Date		Yes
Web Request	status code	HTTP response status code (100-599).		100 - 599	Yes
Web Request	device type	Client device type: Desktop, Mobile or Tablet.		Desktop, Mobile, Tablet	Yes

7. Relationships

Table 6: Relationships

Parent Entity	Child Entity	Relationship	Cardinality	Optionality
Company	Customer	employs	1 : N	A company has 0..N customers (only buyers); a customer has exactly 1 company
Company	Supplier	sells as	1 : 1	Only supplier companies; each has exactly 1 supplier account
Category	Product	classifies	1 : N	Every product has exactly 1 category
Supplier	Supplier Offer	offers	1 : N	A supplier has 0..N offers
Product	Supplier Offer	is offered as	1 : N	A product has 2..5 offers
Customer	Order	places	1 : N	A customer has 0..N orders; an order has exactly 1 customer
Company	Order	is billed for	1 : N	Mandatory; equals the customer's company
Order	Order Line	contains	1 : N	An order has 1..N lines
Supplier Offer	Order Line	is bought in	1 : N	A line buys exactly 1 offer (implies Product and Supplier)
Marketing Lead	Order	converts into	1 : 0..1	Optional on both sides; most orders have no lead
Date	Order	is order date of	1 : N	Mandatory
Date	Marketing Lead	is lead date of	1 : N	Mandatory
Date	Web Request	is request date of	1 : N	Mandatory

8. History & Change Handling

Table 7: History & Change Handling

Entity	Change Type	Approach (SCD)	Reason
Company	Attribute changes (name, type, tax id, rating, location)	SCD Type 2 - new version with validity dates	Orders must be reported with the company attributes valid at order time (BR-12)
Customer	Contact data changes (e-mail, phone, job title)	SCD Type 1 - overwrite	Only the current contact data is relevant for reporting
Supplier / Supplier Offer	Rare changes (price, lead time)	SCD Type 1 - overwrite	No historical supplier analysis required
Product / Category	Deactivation, price changes	SCD Type 1 - overwrite	Order lines keep their own unit price, so product price history is not needed
Order	Status and payment status progress	Overwrite latest state on the fact	KPIs use the current status; status history not required
Marketing Lead	Funnel stage, score, conversion	Overwrite latest state (accumulating fact)	Funnel reporting uses the current stage
Web Request / Order Line / Date	No changes	Insert only	Immutable events / static calendar

Figure 2 shows how history is kept for companies: a changed company has an expired version and a current version.

```

41 SELECT
42     company_id, company_key, rating, city, effective_start_date, effective_end_date, is_current
43 FROM
44     warehouse.dim_companies
45 WHERE company_id IN (
46     SELECT company_id
47     FROM warehouse.dim_companies
48     GROUP BY company_id
49     HAVING COUNT(*) > 1)
50 ORDER BY company_id, effective_start_date
51 LIMIT 12;
52

```

Data Output Messages Notifications

Showing rows: 1 to 10 Page No: 1

	company_id character (32)	company_key [PK] integer	rating numeric (2,1)	city character varying (100)	effective_start_date timestamp without time zone	effective_end_date timestamp without time zone	is_current boolean
1	0178cbb0c71242cdaf28bdb80253e5...	16	2.6	London	2026-02-12 05:14:12	2026-09-30 17:27:45.417458	false
2	0178cbb0c71242cdaf28bdb80253e5...	2502	2.8	Birmingham	2026-09-30 17:27:45.417458	[null]	true
3	3c916deb199f40baabdfcae0b56109...	606	3.1	New York	2025-08-01 03:57:10	2026-09-30 17:27:45.392565	false
4	3c916deb199f40baabdfcae0b56109...	2503	3.1	New York	2026-09-30 17:27:45.392565	[null]	true
5	88245c3e81444b94b9e3c21043c88f...	1373	3.8	Austin	2026-04-18 01:54:49	2026-09-30 16:22:08.695862	false
6	88245c3e81444b94b9e3c21043c88f...	2501	3.7	Seattle	2026-09-30 16:22:08.695862	[null]	true
7	8b4c4251f551408b8ed1a4e0ca2a3f...	1410	4.3	New York	2026-09-08 05:33:19	2026-09-30 17:30:38.062934	false
8	8b4c4251f551408b8ed1a4e0ca2a3f...	2505	4.4	Chicago	2026-09-30 17:30:38.062934	[null]	true
9	b71a61ae2a4249eebf73bb0801429c...	1836	3.4	Seattle	2024-12-12 14:13:54	2026-09-30 17:27:45.430529	false
10	b71a61ae2a4249eebf73bb0801429c...	2504	3.3	New York	2026-09-30 17:27:45.430529	[null]	true

Figure 2: SCD Type 2 — companies with more than one version

9. Business Rules

Table 8: Business Rules

Rule ID	Rule	Entities Affected
BR-1 / BR-2	Customers belong only to Buyer companies; suppliers only to Supplier companies	Company, Customer, Supplier
BR-4	A product is offered by 2-5 suppliers; one offer per supplier-product pair	Product, Supplier, Supplier Offer
BR-5	An order belongs to one customer and its company, and cannot predate the customer	Order, Customer, Company
BR-6 / BR-7	An order has at least one line; order total = sum of line totals; line total = quantity x unit price - discount	Order, Order Line
BR-8	A lead converts into at most one order; conversion status derives from the existence of that order	Marketing Lead, Order
BR-10	Cancelled orders are kept but excluded from revenue	Order, Order Line
BR-12	Company history is preserved (SCD2); facts link to the version valid at event time	Company, Order, Order Line

Rule IDs BR-1 to BR-10 are the business rules observed in the source (02 Source System Analysis, section 6); only the rules that shape the logical model are repeated here. BR-12 is a modelling rule defined in this document.

Figure 3 shows the rows that broke a business rule; they are kept in rejected_records with a reason instead of being dropped.

```

76  SELECT table_name, reject_reason, COUNT(*) AS rejected_rows
77  FROM intermediate.rejected_records
78  GROUP BY table_name, reject_reason
79  ORDER BY table_name, reject_reason;
80

```

Data Output Messages Notifications			
	table_name character varying (50)	reject_reason character varying (100)	rejected_rows bigint
1	marketing_leads	invalid estimated_order_value	741
2	marketing_leads	invalid lead_score	723
3	marketing_leads	missing required field	794
4	web_logs	invalid bytes_sent	1106
5	web_logs	invalid http_method	1133
6	web_logs	invalid status_code	1119
7	web_logs	missing required field	1159

Figure 3: Rows that broke a business rule, kept in rejected_records

10. Logical-to-Physical Mapping

Table 9: Logical-to-Physical Mapping

Logical Entity	Physical Table(s)	Physical Treatment
Date	warehouse.dim_date	Generated calendar; integer key YYYYMMDD
Company	warehouse.dim_companies	SCD2: surrogate company_key, effective dates, is_current, partial unique index on current row
Customer	warehouse.dim_customers	SCD1: surrogate customer_key
Supplier	warehouse.dim_suppliers	SCD1: surrogate supplier_key
Category	warehouse.dim_products (category_id, category_name)	Flattened into Product (no separate table)
Product	warehouse.dim_products	SCD1: surrogate product_key
Supplier Offer	warehouse.dim_supplier_product	SCD1; holds supplier_key and product_key
Order	warehouse.fact_orders	Monthly range partitions on date_key; item_count derived
Order Line	warehouse.fact_order_items	Offer resolved to product_key + supplier_key; order attributes denormalised; partitioned
Marketing Lead	warehouse.fact_leads	order_id attached; partitioned on lead date
Web Request	warehouse.fact_web_logs	Insert-only; partitioned

11. Assumptions, Constraints & Open Issues

Table 10: Assumptions, Constraints & Open Issues

ID	Type	Description	Owner	Status
A1	Assumption	Current-state reporting is sufficient for all entities except Company (SCD Type 2 as required).	Data Engineering	Accepted
C1	Constraint	Marketing Lead and Web Request are not linked to Company / Customer: the sources share no common key.	Data Engineering	Accepted